

MANAGING REACTION WHEEL MICROVIBRATION ON A HIGH RESOLUTION EO SMALL SPACECRAFT

Geert Smet⁽¹⁾, Guy Richardson⁽¹⁾, Simon McLaren⁽¹⁾, Andrew Haslehurst⁽¹⁾

⁽¹⁾ Surrey Satellite Technology Ltd, Tycho House, 20 Stephenson Road, Surrey Research Park, Guildford, GU2 7YE, United Kingdom, Email: g.smet@sstl.co.uk, g.richardson@sstl.co.uk, s.mclaren@sstl.co.uk, a.haslehurst@sstl.co.uk

ABSTRACT

The SSTL 300 S1 spacecraft is capable of capturing images with a spatial resolution of 1m. One of the main technical challenges for this mission is to eliminate effects of microvibration on the imagery. Microvibration on the SSTL 300 S1 spacecraft is mainly caused by mechanical noise generated by the reaction wheels. This has been addressed by optimising the reaction wheel's microvibration performance. The optimisation is driven by the modes at which the spacecraft is sensitive to microvibration. Furthermore, a new isolation system has been designed to further reduce the reaction wheel noise levels. Since the isolation system has very low resonant frequencies the wheels need to be locked down to meet the structural requirements during launch. This paper presents the reaction wheel optimisation and the design and test results of the integrated wheel isolation and lockdown system.

1. INTRODUCTION

Microvibration is a low level disturbance, which is exceptionally difficult to be controlled or reduced by the attitude and orbital control system (AOCS) of a spacecraft. Mechanisms, such as the four 100SP-O reaction wheels or two antenna pointing mechanisms (APMs) on the SSTL 300 S1 spacecraft, are the most common sources of microvibration. Microvibration causes problems for sensitive payloads, such as the 1m high resolution camera on the SSTL 300 S1 spacecraft. It can be the cause of modulation transfer function (MTF) reduction, blurring of the image, or can generate artefacts in the imagery. Fig. 1 shows an example of these artefacts.

Microvibration is a system level problem. It can be handled in many different ways on a spacecraft, such as:

- The payload can be designed to be less sensitive to microvibration or be mechanically isolated from the rest of the spacecraft.
- The structure of the spacecraft can be designed to reduce the microvibration transmissibility.
- It is conceivable to reduce some of the effects of microvibration by post-processing imagery.
- Microvibration can be managed by reducing the mechanical noise generated by the noise sources or by isolating the noise sources.



Figure 1. Image with microvibration artefacts.

SSTL has a distinct advantage managing microvibration, since the payload, structure and mechanisms are all designed in house. A conscious choice was made to focus the efforts predominantly on reducing the microvibration generated by the noise sources. This avoids further technical constraints on the design of the high resolution imager. The consequence of this decision is the need to redesign the existing 100SP-O reaction wheel and to develop of a new reaction wheel isolation system.

As well as allowing for a strategic decision on how to handle microvibration, owning the entire system design gives SSTL detailed knowledge on how exactly the spacecraft is sensitive to microvibration. It would not be straightforward to receive the necessary information if a crucial part of the system, such as the payload, was acquired externally. The SSTL 300 S1 spacecraft is sensitive to microvibration at several frequencies between 50Hz and 250Hz. This allows for a clever modal design of the noise sources, tailoring the mitigation approach to minimise noise generated in this frequency range specifically.

2. MICROVIBRATION OPTIMISATION OF THE 100SP-O REACTION WHEEL

2.1. Reaction wheel noise sources

An excellent description of noises sources inside reaction wheels is available in [1]. A very short summary is given here. There are three noise sources within reaction wheels:

- Wheel unbalance
- Motor noise
- Bearing disturbances

Each of these noise sources is speed dependent. With the exception of wheel unbalance, the disturbances are generally small in amplitude. However, when they excite wheel structural modes they can be amplified to a multiple of their original amplitude.

For the wheel structural modes, the terminology from [1] will be adopted here. Typically the three lowest modes are relevant. For the 100SP-O reaction wheel these modes are:

- Ax - Axial translational mode of the inertia disk
- Ra - Antiparallel rocking mode of the inertia disk, rocking of the disk is opposite to the bearing shaft
- Rp - Parallel rocking mode of the inertia disk, rocking of the disk is parallel to the bearing shaft.

Throughout this paper Campbell plots will be used to present wheel data. These plots present the power spectral density (PSD) of the wheel versus speed and frequency. As pointed out in [1], using PSDs has the disadvantage that the presented data is dependent on the filter bandwidth used during the fast Fourier transformation (FFT). In this paper, the Campbell plots are used to show relative changes before and after changes to the wheel and to show the performance of the isolation system. Furthermore, Campbell plots are a good visual tool where all speed dependent noise sources show up as diagonals and structural resonances show of as horizontal lines, with the exception of Rp which splits up in two whirl modes due to the gyroscopic effect. Rather than showing all 6 force components, this paper will use only Fy and Fz since all structural modes are represented in these components.

2.2. 100SP-O reaction wheel description

The 100SP-O reaction wheels are designed for a 3-axis control system, in momentum bias or zero bias mode. The 100SP-O wheels are configured to provide a highly agile attitude control solution for Earth Observation missions. Tab. 1 shows the 100SP-O specifications.

In contrast to SSTL's successful line of smaller 10SP wheels, the 100SP-O wheels are oil lubricated. This provides a more stable microvibration noise signature throughout the life of the wheel.

Table 1. 100SP-O wheel specifications.

	100SP-O
Mass	2.6 kg
Angular momentum	1.5 Nms
Max torque	0.11 Nm
Dimensions	Ø 120 x 120 mm
Design lifetime	7+ years
Operating temperature	-20°C to +50° C
Power (at max speed of 5000 rpm at 20° C)	~10 W

2.3. Reaction wheel microvibration optimisation approach

Rather than designing a reaction wheel specifically for a low microvibration signature, SSTL has elected to optimise the existing qualified 100SP-O wheel in order to reduce the risks in when qualifying the optimised wheel. The approach in optimising the 100SP-O reaction wheel behaviour is threefold.

Firstly, the aim is to move the reaction wheel structural modes away from the microvibration sensitive modes. The Campbell plot in Fig. 3 shows the behaviour of the original 100SP-O wheel. Both the axial translation mode Ax at ~270Hz and the parallel rocking mode Rp at ~260Hz are close to the microvibration sensitive frequencies at 250Hz. The antiparallel mode Ra at 720Hz is sufficiently high.

Secondly, the noise sources in the wheel exciting these structural modes should be minimised. In order not to affect the qualification of the reaction wheel motor, lowering the motor noise has not been considered. Only the bearing noise and the wheel unbalance will be lowered where possible.

Finally, the noise sources in the reaction wheel are all speed dependent, so the reaction wheel speeds as used by the AOCs can be tuned or limited in order to optimise the overall microvibration performance.

The approach for this optimisation exercise has been predominantly experimental. A dedicated test wheel was built and tested on the SSTL Kistler table (Fig. 2) in the following configurations with their attributed objectives:

- Internal damping introduced inside the wheel: Reduce amplitude of noise generated.
- Light weight inertia disk: Shift structural modes.
- Hybrid ceramic bearings: Reduce amplitude of noise generated.
- Hybrid ceramic bearings and increased bearing preload: Shift structural modes.

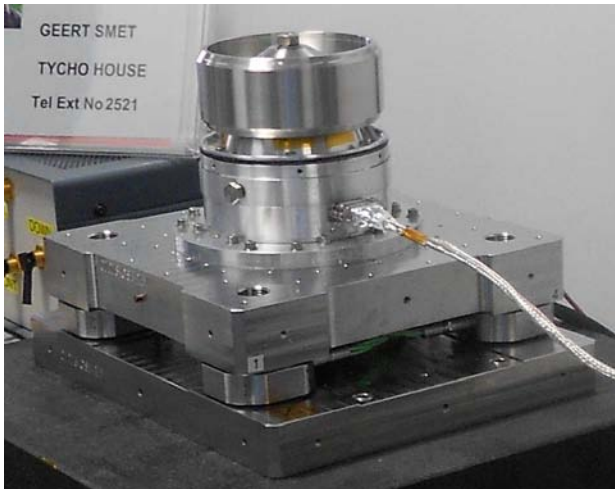


Figure 2. 100SP-O wheel without its cover on the SSTL Kistler table.

2.4. Internal damping

Two separate methods of introducing damping inside the reaction wheels have been tested:

- Visco-elastic washers in the load path between the wheel motor and the wheel housing.
- Replacing a structural component in the load path with a high damping boron carbide in aluminium alloy metal matrix composite.

The wheel configuration with the visco-elastic washers in the load path generated an improvement over a limited portion of the frequency range. Between 200 and 300Hz an improvement of up to an order of magnitude is measured. The introduction of the washers also had an influence on one of the wheel modes. Ra drops from 720Hz to 560Hz.

Implementing the visco-elastic washers has been abandoned due to drawbacks of this configuration. The conductive thermal link between the motor and the rest of the wheel is broken, which requires a more elaborate design change to rectify. Furthermore, there are serious concerns regarding the cleanliness on the inside of the wheel when introducing these washers.

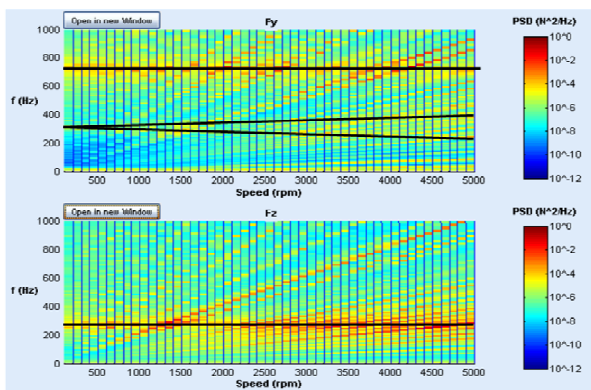


Figure 3. 100SP-O Fy & Fz Campbell plot.

They are difficult to clean and potentially generate particles. There is a risk that particles inside the wheel would adversely affect the wheel's lifetime and could ultimately lead to premature wheel failure.

Replacing a structural component with an identical component made out of boron carbide in aluminium alloy metal matrix composite only yields marginal improvements. Small reductions in noise are found in the 300Hz region. This change was not retained in the final wheel design, since the additional cost does not outweigh the performance increase. It needs to be noted that the component shape and form was not optimised in any way to make full use of the material's damping capabilities, so better performance may be possible.

2.5. Light weight inertia disk

In this configuration the stainless steel inertia disk was replaced by an aluminium inertia disk. This reduces the disk mass from 1.075kg to 0.363kg and the inertia from 1.39Nms to 0.47Nms. This significantly impacts the reaction wheel performance and is not viable to fly on the SSTL 300 S1 spacecraft. It is intended to quantify the potential for changing the wheel's structural modes.

Rp moves from 260Hz to 450Hz, Ra moves from 720Hz to 1000Hz and Ax moves from 270Hz to 450Hz. The former is due to the reduction in inertia, whereas the latter two are due to the reduction in mass of the inertia disk. Reducing the mass of the inertia disk without reducing its inertia would move Ax away from the microvibration sensitive modes of the spacecraft.

A significant move requires a big reduction in mass of the inertia disk. This could only be achieved by a composite disk design. The reduction in mass should also be executed without a significant loss in stiffness of the disk itself. Due to the magnitude of the design exercise to optimise the disk, an optimised disk has not been included in the final wheel design. It is one of the potential risk mitigation options should the performance of the entire system be worse than expected.

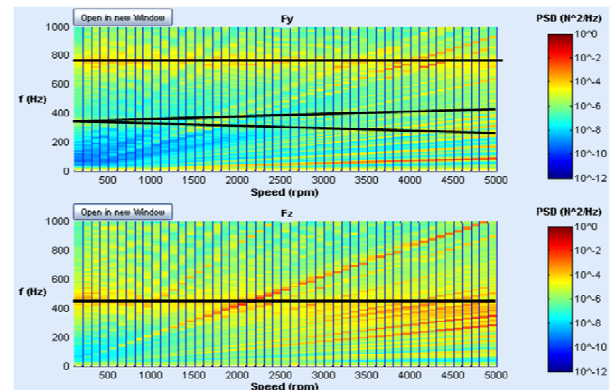


Figure 4. 100SP-OC Fy & Fz Campbell plot.

2.6. Hybrid ceramic bearings

The 100SP-O wheel uses bearings with stainless steel balls in stainless steel bearing races. Hybrid ceramic bearings have ceramic balls in stainless steel races. The main advantage of hybrid ceramic bearings is that they exhibit a much smoother surface finish and the balls are 60% lighter than steel balls. This reduces microvibration disturbances. Additionally the removal of asperity contacts from cold welding is eliminated due to the dissimilar materials, allowing the smooth surfaces to be maintained compared to steel on steel. This gives greater confidence that microvibration requirements can be maintained over the mission life.

Changing the bearings requires the new wheel to be qualified. Regardless, this configuration was tested as the bearing manufacturer predicted a reduction of the mechanical noise of two to seven orders of magnitude.

Even though the ceramic balls change the bearing stiffness, no change was seen in the wheel structural modes. The bearing disturbances are lower. One order of magnitude reduction of the noise is measured in the Rp 200-300Hz peaks. Similarly, one order of magnitude reduction is measured in the 700Hz Ra peak. Most significantly, two orders of magnitude reduction of the mechanical noise is measured in the Ax 200-300Hz peaks. Due to these improvements, hybrid ceramic bearings have been included in the final wheel design.

2.7. Hybrid ceramic bearings with increased preload

Increasing the preload of the bearings changes the structural behaviour of the wheel. Since at the time of test it was already apparent hybrid ceramic bearings significantly improved the microvibration performance, the nearly doubling of the preload was tested in conjunction with these bearings.

Rp moves from 260Hz to 350Hz, Ra moves from 720Hz to 750Hz and Ax moves from 270Hz to 400-450Hz. Especially the latter has a big impact on the microvibration performance of the reaction wheel. There is a reduction of two orders of magnitude in the Rp 200-300Hz peaks, one order of magnitude in the Ra 700Hz peak and up to 4 orders of magnitude in Ax 200-300Hz peaks, although it is more accurate to say the peak shifts to 400-500Hz with 2 orders of magnitude reduction. These results merit the inclusion of a higher preload in the final wheel design.

2.8. Conclusion

All of the investigated configurations to optimise the wheel microvibration performance have their merits. The final design trade off favoured the use of hybrid ceramic bearings and a higher preload.

In addition, the wheel balance requirements for this spacecraft have been tightened. Finally, by reducing the maximum wheel speed during imaging to 2000rpm, a significant performance increase has been made. Fig. 4 shows a Campbell plot of the hybrid ceramic bearing wheel design named 100SP-OC, highlighting the shift in structural modes. The 100SP-OC wheel has completed its qualification program and is currently in life test.

A drawback of the wheel redesign is a reduction in the vibration loads it can withstand. However, as this wheel is optimised for microvibration performance, it always needs to be carefully protected from the launch loads.

3. WHEEL ISOLATION AND LOCKDOWN SYSTEM

3.1. Design rationale

Further microvibration reductions can be made by isolating the wheels from the rest of the spacecraft. SSTL has ample experience designing isolation mounts for their reaction wheels. Typically, visco-elastic spring elements are used to isolate one wheel. These isolation mounts have the dual purpose to protect the spacecraft from microvibration generated by the wheel as well as protecting the wheel from the vibrations during launch. The latter in turn improves the wheel microvibration performance. Literature [1] confirms that improving the mechanical environment of the wheel during launch can yield significant improvements of its microvibration performance.

Previous SSTL isolation mounts had resonant frequencies of 100Hz and 60Hz. For the latter, the dynamic envelope of the wheel during launch became too big to fit in the platform. It required the introduction of soft mechanical end stops, snubbers, to limit the dynamic envelope of the wheel. The impact events when contact is made with the snubbers result in a harsher vibration environment for the wheel.

For the SSTL 300 S1 spacecraft, the frequencies of previous isolation systems could have been problematic since the spacecraft has microvibration sensitive modes at lower frequencies. Consequently, the resonant frequency of the isolation system is lowered below the spacecraft microvibration sensitive frequency at 50Hz. This also provides more attenuation at the higher frequencies. The downside is a bigger dynamic envelope, although the available space in the SSTL 300 S1 spacecraft for the reaction wheels is limited. This requires either a snubber system, with its known drawbacks for the microvibration performance of the wheel, or locking the wheels down during launch. Hard locking the wheels during launch would also be detrimental for their microvibration performance. The design therefore includes a 'soft lock' system which provides attenuation for the wheels during launch.

In order to lower the resonant frequency of the isolation systems, two different approaches are viable. The spring elements can be made more compliant and the isolated mass can be increased. Whilst efforts have been made to select more compliant springs, the preference was to build on previous heritage with visco-elastic spring elements. Therefore, less spring elements are used to isolate a larger mass. The original idea to isolate all four reaction wheels together proved impractical because the available space did not lend itself to this solution and it is difficult to make the structure holding all the reaction wheels sufficiently stiff. The final selected configuration isolates the reaction wheels in pairs.

A last change from SSTL's heritage isolation mounts is the thermal solution. The visco-elastic elements used are poor thermal conductors and copper thermal straps are used to create a conductive thermal link. Their stiffness is detrimental to the microvibration performance. On the SSTL 300 S1 spacecraft, the wheels have direct access to cold space. Each pair of wheels has been given a radiator panel passively tuned to keep them within their operational temperature range in order to eliminate the thermal straps and improve microvibration performance.

In summary, the wheel isolation and lockdown system has four functions:

- It isolates the spacecraft from microvibration generated by the reaction wheels.
- It locks down the reaction wheels during launch.
- It protects the reaction wheels from the launch loads when it is in its locked down configuration.
- It passively self regulates the temperature of each pair of reaction wheels as a stand alone system.

3.2. Design description

Fig. 5 shows a schematic overview of the design and its components. Each pair of wheels is held together in a carbon fibre brace. The braces are connected to a honeycomb panel, the Mu-panel. The entire assembly in turn is connected to the space facing facet (SFF) of the spacecraft structure. Fig. 7 shows the hardware.

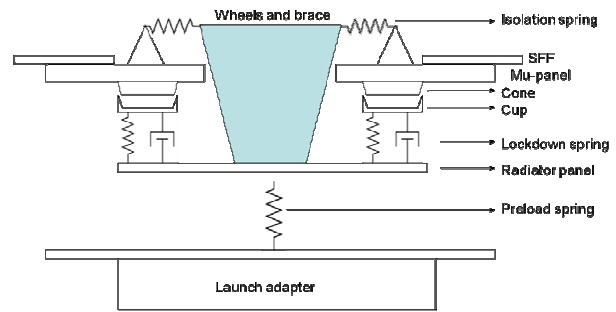


Figure 5. Schematic overview of the wheel isolation and lockdown system.

The Mu-panel is designed for stiffness and includes a stiffener. Its modal design is optimised to avoid coupling with spacecraft structure. The Mu-panel is connected to the SFF where it is in turn connected to the central thrust tube (Fig. 6).

There are three sets of springs in this design, each of them with their own purpose: isolation springs, lockdown springs and preload springs

Four isolation springs connect a pair of wheels in their brace to the Mu-panel. These springs are the only connection between the wheels and the spacecraft after wheel deployment. The isolation springs act as a low pass filter for mechanical noise generated by the reaction wheels. The first natural frequency of the system in its deployed configuration is around 10Hz.



Figure 6. Wheel isolation and lockdown system installed in the spacecraft (internal view).

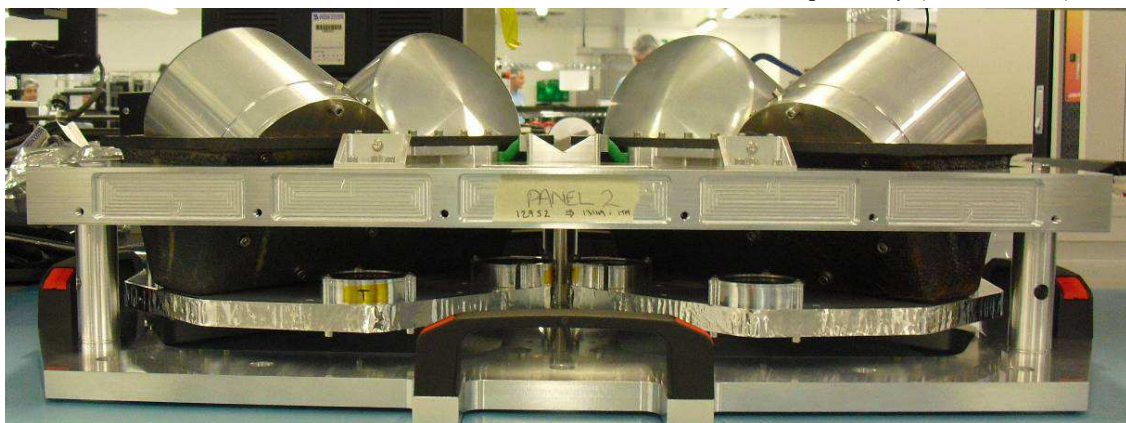


Figure 7. Wheel isolation and lockdown system on its carrier. Mu-panel stiffener removed.

The lockdown springs sit underneath the cup cone interface which is installed between the radiator panel and the Mu-panel (Fig. 8). Each pair of wheels has four cups and cones with a lockdown spring each. In the locked down position of this system the cups and cones will be engaged. The purpose of the lockdown springs is to make sure the wheels have some attenuation from the launch loads when they are locked down. This minimises the impact of the launch loads on the microvibration behaviour of the reaction wheels. The system in its locked down configuration has a natural frequency of around 70Hz.



Figure 8. Cup-cones. System in deployed state.

The preload springs are connected to the launch adapter and provide preload to the system in its locked down state (Fig. 9). Each spring interfaces with a radiator panel but is not permanently connected to it. The preload springs are compressed during installation of the spacecraft on the launch adapter using the mass of the spacecraft. Upon spacecraft separation from the launcher, the spacecraft will move away from these preload springs and the wheels deploy. This avoids the need for an active hold down mechanism.

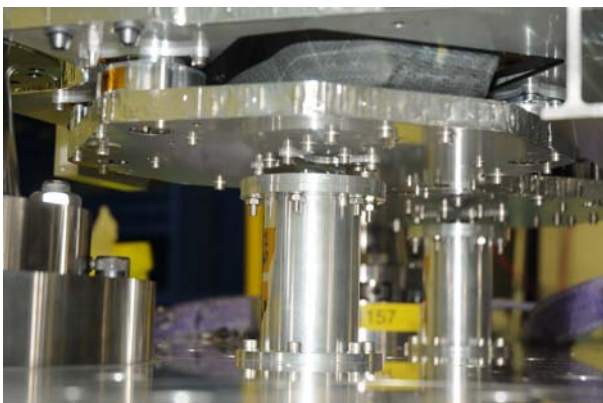


Figure 9. Preload springs. System in locked state.

3.3. Wheel isolation system development

The brace for the wheel isolation system is made out of carbon fibre reinforced plastic (CFRP). The shape of the brace lends itself to this manufacturing process. In order to verify the concept, a breadboard model was manufactured.

The breadboard model was tested on the Kistler table (Fig. 10). The goals of these breadboard tests are:

- To verify the optimal position of the isolation springs as predicted through analysis
- Test different isolation spring materials
- Confirm the concept of isolating two wheels together, specifically the effect of two wheels running at different speeds
- To verify the design of the brace holding the two wheels and assess its impact on the performance

The isolation springs used have the same geometry as used in previous isolation mounts even though this geometry is not optimised for this configuration. Many different locations and orientations have been analysed. The purpose was to tune the first six rigid body modes of the system close together and as low as possible. Tests on the breadboard model confirm the analysis and favour a symmetric set-up using four isolation springs.

Three different materials have been tested for the isolation springs. The selected material performs one order of magnitude better than the other two materials in this configuration and has been selected for the design.

Whilst isolating two wheels together is advantageous to increase the isolated mass and reduce the isolation system resonant frequency, it is a less predictable system. The wheels could interact at certain speeds and the noise could be amplified. In order to test this, extensive testing has been performed with one wheel going through its speed range and the other wheel held at different constant speeds, as well as stepping both wheels through the speed range together. No unexpected behaviour has been measured and the wheels stepping through the speed range together was identified as the worst case performance.

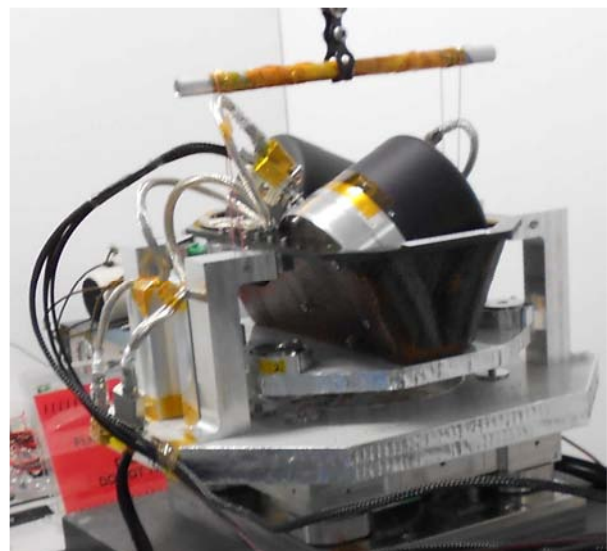


Figure 10. Two 100SP-OC wheels in their brace on the SSTL Kistler table

Since the gravity sag on the compliant isolation springs is significant, the system was gravity offloaded during the Kistler tests. Test results without gravity offloading differ little from results with gravity offloading. For spacecraft level tests, gravity offloading is not possible but tests without gravity offloading are valid.

The isolation system significantly reduces the noise of the reaction wheels between 50 and 200Hz. However, between 250 and 400Hz, much more noise than expected was measured. This was attributed to modes of the wheels in the brace. The brace holding the wheels needs to be stiffer.

A second breadboard brace has been manufactured with an increased thickness and a stiffer material. The overall performance of this system shows a noise reduction of more than two orders of magnitude over most of the frequency range when compared to a hard mounted 100SP-OC reaction wheel. A mode at 350Hz in F_y is still prominent in the results.

A further improvement was made for the final brace design by further increasing the overall thickness, focussing on the corners to increase the frequency of the modes deemed responsible for the 350Hz peak. The Kistler tests of the third version of the brace did not yield the expected improvement. On further investigation, it became clear that the mechanical ground support equipment (MGSE) that connects the isolation system to the Kistler table has modes at that frequency. Modified MGSE yielded the expected performance increase. Fig. 11 shows the Campbell plot of a pair of isolated wheels stepping through the speed range together. The results are reliable up to 500Hz. Measured microvibration barely breaches the Kistler table noise floor. The noise floor for F_z at $\sim 10^{-8} \text{ N}^2/\text{Hz}$ is higher than for F_y at $\sim 10^{-10} \text{ N}^2/\text{Hz}$. Artefacts at 50Hz increments are due to electrical background noise.

Spacecraft level tests are currently ongoing to verify the performance at spacecraft level. The focus of these tests is on the modal behaviour of the Mu-panel.

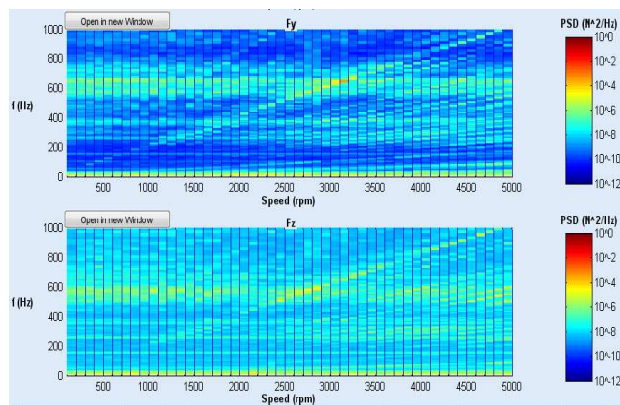


Figure 11. 2x100SP-OC isolated F_y & F_z Campbell plot.

3.4. Lockdown system development and qualification

The following aspects of the lockdown system development and qualification will be discussed here:

- Lockdown system modal design
- Impact on spacecraft separation
- Cup-cone design and verification
- Sine vibration and acoustic tests on the spacecraft structural qualification model (SQM)

The reaction wheels are locked down in order to withstand the launch loads. In locked down state, the lockdown springs still attenuate the vibration loads going into the wheel. The system is tuned to with its modes sufficiently low to provide protection to the wheels, sufficiently high to limit the dynamic envelope of the wheels and their brace and tuned away from the main spacecraft modes to avoid coupling. The first natural frequency of the system is 70Hz. The stiffness of the system is mainly generated by the lockdown springs, since the preload springs are very compliant springs.

The lockdown system impacts the separation velocity and spacecraft tip-off rate since it introduces additional energy into the separation. All three types of spring elements affect the separation. In order to minimise this, the potential energy stored in the springs has been minimised. Each preload spring provides 2kN preload, but only has a stroke of 1mm and will remain preloaded at 1.95kN. The total energy introduced into the separation is less than 7J for all springs combined.

The cup-cone geometry needs to be such that the launch loads can be taken without gapping and deployment is guaranteed. The cup-cones are assumed frictionless when calculating their load bearing capabilities. The preload is limited to the spacecraft mass due to the design choices made for the preload springs. The preload used in the calculations is lower still, since the static acceleration of the launch vehicle will reduce the effective preload. The load case driving the cup-cone geometry is dominated by the moment loads. Due to the limited available space between the Mu-panel and radiators, this results in a steep half cone angle.

This steep angle means that the friction coefficient between the cup and cone needs to be low to avoid self locking cup-cones. This has driven the material choice for the cone made from a self-lubricated plastic and the cup made from aluminium coated in an electroless nickel and polytetrafluoroethylene (PTFE) coating. Upon spacecraft separation, the cup-cones will be pushed apart by the isolation springs and the inertia of the wheels in their brace. Abstraction is made of the latter since the separation acceleration is currently unknown. The springs provide 100N of force per wheel brace. Applying safety factors, the adhesion force per cup-cone pair needs to be below 3.33N.

All cup-cone pairs are compressed to their maximum preload and the force to pull them apart is measured. Dedicated tests have been introduced to assess the effect of creep and thermal expansion. No cup-cone pair has needed a force higher than 0.5N to be separated for any of the flight hardware. Some of the cups and cones made for the SQM hardware did require forces of up to 15N. This non-compliance is attributed to cup-cone pairs which were not properly match machined. These cup-cones have been intentionally installed on the SQM, in order to assess whether this would impact successful deployment after the vibration and acoustic tests.

Sine vibration and acoustic tests on the spacecraft SQM verify the lockdown system's modal design, the lockdown system's mechanical integrity and measure the vibration inputs to the reaction wheels. There is no evidence of hammering between the cups and cones during either vibration or acoustic tests. This means the cup-cone geometry and preload are sufficient to take the loads. The modal design has been verified. A clear mode at 69.5Hz has been measured during sine tests. During sine the peak response at wheel disk was acceptable. During acoustic qualification test very low vibration levels have been measured on the wheel disk. The microvibration performance of the wheel will not be affected significantly at these levels. A successful deployment test was performed after both the vibrate and acoustic test, even though some of the out of spec cup-cone pairs were installed. The only outstanding test for the lockdown system is the SQM level shock test.

4. CONCLUSIONS

The microvibration optimisation of the 100SP-O reaction wheel has resulted in the 100SP-OC wheel with hybrid ceramic bearings and a higher bearing preload. This wheel has lower noise due to bearing disturbances and has had its structural modes tuned away from the spacecraft microvibration sensitive frequencies. Further improvements have been made by tightening the balancing requirements and by reducing the operational speed range during imaging.

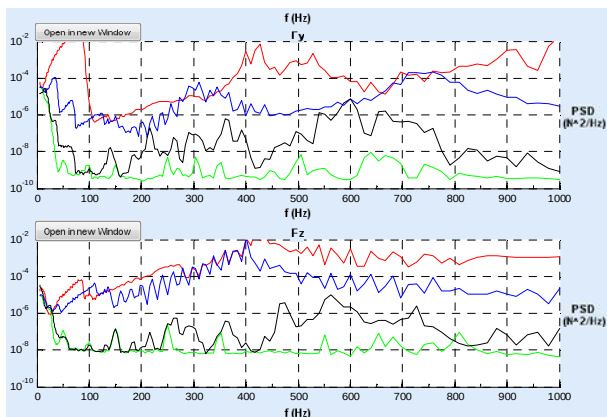


Figure 12. Peakhold comparison wheel performance.

Changing the disk design and including internal damping in the wheel as means of further improving the reaction wheel microvibration performance have been assessed. Even though they were not retained in the 100SP-OC design, they merit further investigation especially for inclusion in a new wheel design.

A new isolation system has been designed which isolates the reaction wheels in pairs. Its natural frequency is very low in order to provide attenuation at all microvibration sensitive frequencies of the spacecraft. The resulting noise has very little power at the spacecraft sensitive frequencies. For large sections of the frequency range it is in fact not possible to measure the performance of the system, since the measured microvibration is dominated by the background noise.

Fig. 12 shows a peakhold plot which is equivalent to the maximum noise measured at any given frequency over a range of speeds. The red curve shows the 100SP-O wheel for a speed range of 100-5000rpm. The blue curve is the 100SP-OC wheel for a speed range of 100-2000rpm. The black curve is two isolated 100SP-OC wheels for a speed range of 100-2000rpm. The green curve is the background noise on the Kistler table. The resulting microvibration solution performs several orders of magnitude better than the original wheel over the entire frequency range and performs very well at the spacecraft microvibration sensitive frequencies.

A novel soft lockdown system for the reaction wheels has been developed. The wheels are locked down directly from the launch vehicle and deploy on spacecraft separation. This design avoids the need for an active hold down mechanism and attenuates the launch loads. The design has successfully passed all its tests.

Sixteen flight reaction wheels and four isolation and lockdown systems are currently in build and will be delivered before the end of 2013.

5. REFERENCES

1. Heimel, H. (2011). Spacewheel Microvibration – Sources, Appearance, Countermeasures. In Proc. GNC2011 8th International ESA Conference on Guidance, Navigation and Control Systems.

6. ACKNOWLEDGEMENTS

The authors gratefully acknowledge the contributions of: Paul Carter for providing the original idea of locking the reaction wheels from the launch vehicle, Zeger De Groot for his contributions to the approach to managing microvibration on this spacecraft, Brian Lepage for building and commissioning the new SSTL Kistler table, Alizee Malavart for writing the software to process and present all the Kistler test data and Tom Clatworthy for getting the thermal solution for the reaction wheels to work.